

A Two-Distribution Compounded Statistical Model for Radar HRRP Target Recognition

Lan Du, Hongwei Liu, *Member, IEEE*, Zheng Bao, *Senior Member, IEEE*, and Junying Zhang, *Member, IEEE*

Abstract—In the statistical target recognition based on radar high-resolution range profile (HRRP), two challenging tasks are how to deal with the target-aspect, time-shift, and amplitude-scale sensitivity of HRRP and how to accurately describe HRRPs statistical characteristics. In this paper, based on the scattering center model, range cells are classified, in accordance with the number of predominant scatterers in each cell, into three statistical types. After resolving the three sensitivity problems, this paper develops a statistical model comprising two distribution forms, i.e., Gamma distribution and Gaussian mixture distribution, to model echoes of different types of range cells as the corresponding distribution forms. Determination of the type of a range cell is achieved by using the rival penalized competitive learning (RPCL) algorithm, while estimation for the parameters of Gamma distribution and Gaussian mixture distribution by the maximum likelihood (ML) method and the expectation-maximization (EM) algorithm, respectively. Experimental results for measured data show that the proposed statistical model not only has better recognition performance but also is more robust to noises than the two existing statistical models, i.e., Gaussian model and Gamma model.

Index Terms—A two-distribution compounded statistical model, expectation-maximization (EM) algorithm, high-resolution range profile (HRRP), radar automatic target recognition (RATR), rival penalized competitive learning (RPCL).

I. INTRODUCTION

A HIGH-RESOLUTION range profile (HRRP) is the amplitude of the coherent summations of the complex time returns from target scatterers in each range cell, which represents the projection of the complex returned echoes from the target scattering centers onto the radar line-of-sight (LOS). It contains the target structure signatures, such as target size, scatterer distribution, etc., thereby radar HRRP target recognition has received intensive attention from the radar automatic target recognition (RATR) community [1]–[19]. References [8] and [9] discussed the properties of target HRRP; [10]–[12] studied

some time-shift invariant features of HRRP; [13] discussed the variation of recognition performance with signal bandwidth; [14] proposed a new feature selection method for HRRP; [15] applied hidden Markov model (HMM) to radar HRRP target recognition, and [16] and [17] proposed some advanced classifiers for HRRP-based RATR. However, there has been little work in the field of radar HRRP-based statistical recognition. By statistical recognition we mean that the class membership of a measurement HRRP $\mathbf{x}$ is determined based on the posterior probabilities of all class memberships ($p(k | \mathbf{x})$, $k = 1, 2, \dots, K$ denotes class membership). Several works [18], [19] showed that statistical recognition has the advantages of giving a measure of confidence for the class predictions, and the opportunity to combine the probabilities with additional information, such as intelligence reports, or possibly other sensor data. Furthermore, since a user needs to decide upon a course of action after classifying a target, the probabilities can be incorporated into a multilevel model that reflects the whole decision making process. For instance, by considering the expected posterior loss of decisions, the different costs involved in making a classification can be taken into account.

Several issues have to be considered when HRRP is applied to statistical recognition. The first one is the well-known target-aspect sensitivity of HRRP. The scattering center model is widely used by high range resolution radar imaging community, which has been proved to be a simple and efficient target model in synthetic aperture radar (SAR) imaging and inverse synthetic aperture radar (ISAR) imaging [20], [21]. According to this model, the variation of target-aspect will lead to scatterers' motion through range cells (MTRC), therefore, the statistical characteristics of target HRRP will be changed substantially. The second one is the time-shift sensitivity of HRRP. In order to decrease the computation complexity, an HRRP is only a part of received radar echo extracted by a range window, in which target signal is included. Thus the position of the target signal in an HRRP can vary with the measurement. The last one is the amplitude-scale sensitivity of HRRP. This comes from the fact that the intensity of an HRRP is a function of radar transmitting power, target distance, radar antenna gain, radar receiver gain, radar system losses and so on. Thus, HRRPs measured by different radars or under different conditions will have different amplitude-scales. Therefore, it is a prerequisite of HRRP-based statistical recognition to deal with the three sensitivity problems of HRRP. In this paper, we make a detailed analysis of the effect of the sensitivity problems of HRRP on statistical recognition, and propose our corresponding solutions.

According to Bayes theorem, we have $p(k | \mathbf{x}) \propto p(\mathbf{x} | k)P(k)$, where $p(\mathbf{x} | k)$ is the class-conditional probability

Manuscript received December 28, 2004; revised July 15, 2005. This work was supported in part by the National Science Foundation of China (NO.60302009) and by the National Defense Advanced Research Foundation of China (No. 413070501). The associate editor coordinating the review of this manuscript and approving it for publication was Prof. Yuri I. Abramovich.

L. Du and Z. Bao are with the National Laboratory of Radar Signal Processing, Xidian University, Xi'an, China (e-mail: dulan@mail.xidian.edu.cn).

H. Liu was with the Department of Electrical and Computer Engineering, Duke University, Durham, NC 27708 USA. He is now with the National Laboratory of Radar Signal Processing, Xidian University, Xi'an, China.

J. Zhang was with the Department of Electrical Engineering and Computer Science, the Catholic University, Washington, DC, USA. She is now with the National Laboratory of Radar Signal Processing, Xidian University, Xi'an, China, and also with the School of Computer Science and Engineering, Xidian University, Xi'an, China.

Digital Object Identifier 10.1109/TSP.2006.873534

density and $P(k)$ is the prior class probability. Thus, estimation of the posterior probability is turned into estimation of the class-conditional density. For an HRRP including many range cells, nonparametric methods of density estimation, e.g., Parzen window method, are impractical due to the huge storage requirement. An alternative approach is to trade flexibility for robustness and to use some simple parametric form for target probability density. There exist two statistical models for target HRRP, i.e., Gaussian model [5] and Gamma model [18], [19]. According to the scattering center model, range cells of target HRRPs are classified into three statistical types, of which the first and second types of echoes can be modeled with Gamma distribution, while the third with Gaussian mixture distribution. The type of a range cell is determined by the number of clusters of the training data, which reflects the number of predominant scatters in the cell. We use the rival penalized competitive learning (RPCL) algorithm [22] to distinguish the type of each cell, due to its advantage of automatic selection of the appropriate number of clusters. Once the type of each range cell is known, echoes of different types of range cells can be modeled as the corresponding distribution forms. Thus, this paper develops a new statistical model comprising two distribution forms, i.e., Gamma distribution and Gaussian mixture distribution, for target HRRP. Parameter estimation of Gamma distribution and Gaussian mixture distribution is achieved by using the maximum likelihood (ML) method and the expectation-maximization (EM) algorithm, respectively. In the recognition experiment based on measured data, the proposed statistical model not only has better recognition performance but also is more robust to noises than Gaussian model and Gamma model.

This paper is organized as follows. In Section II, the problem of how to deal with the target-aspect, time-shift, and amplitude-scale sensitivity of HRRP is discussed. In Section III, the two-distribution compounded statistical model approach to HRRP-based statistical recognition is proposed. In Section IV, the parameter estimation procedure of the proposed statistical model is derived, which includes the estimation for distribution form and distribution parameters. In Section V, the main procedure of radar HRRP target recognition based on the two-distribution compounded statistical model is described. In Section VI, recognition experiments based on measured data are performed to evaluate the recognition performances of Gaussian model, Gamma model and the proposed statistical model. Finally, conclusions are made in Section VII.

II. TARGET-ASPECT, TIME-SHIFT, AND AMPLITUDE-SCALE SENSITIVITY OF HRRP

A. Target-Aspect Sensitivity

Generally, since the targets and their components are much larger than the wavelength of microwave radar, the electromagnetism characteristics of the targets can be described by the scattering center model [20], [21]. According to this model, for high resolution radar system, an aircraft-like target does not appear as a “point target” any longer, but consists of many scatterers separated in several range cells along the radar LOS. Thus different targets have different scattering center models. Furthermore, the scatterer distribution of a target will vary with the target-aspect,

because the variation of target-aspect causes some scatterers to remove from one range cell to another, which is referred to as scatterers’ MTRC. The limitation of target-aspect change for avoiding scatterers’ MTRC is given by [9], [20], [21]

$$\delta\varphi \leq (\delta\varphi)_{\text{MTRC}} \stackrel{\text{Def}}{=} \frac{C}{2BL_x} \quad (1)$$

where B , L_x , and C are the bandwidth of the radar signal after range compression, the maximum target dimension in cross range and the speed of light, respectively. Since the scatterers are localized immovably in each range cell within $(\delta\varphi)_{\text{MTRC}}$, the scattering center model almost remains unchanged. Theoretically, a subset of HRRPs from a target-aspect sector without scatterers’ MTRC, namely, within $(\delta\varphi)_{\text{MTRC}}$, is defined to be an HRRP frame. Nevertheless, this is too strict in application. Generally, we divided the HRRP frame according to the condition that we allow MTRC to occur for the scatterers at the edges of the target and with weak energy, but not for most of the scatterers in one range cell. In this case, $(\delta\varphi)_{\text{MTRC}}$ can be further increased. Thus, an HRRP frame has a corresponding form of statistical characteristics. References [18] and [19] proposed a mixture model approach to incorporate different statistical characteristics of a target from different target-aspect sectors. However, there are several problems for the mixture model approach.

- The recognition performance of a mixture model bears on the number of mixture components. It is difficult to determine the optimal number of mixture components for a mixture model, the point also referred in [18] and [19].
- The time-shift compensation and parameter estimation (detailed illustration shown in Sections II-B and IV) for all training samples from 0–360° are impractical because of the huge computation burden. Usually, the mixture model is trained with a small number of training samples selected from a large aspect area. Nevertheless, the decrease of training samples will degrade the performance of parameter estimation, because the training data set available for HRRP-based RATR is incomplete.
- Target-aspect information of a test sample is difficult to be utilized to further improve the recognition performance for a mixture model even though [8] showed that using the target aspect information can effectively improve recognition performance.

Therefore, this paper does not adopt the mixture model approach to deal with the target-aspect sensitivity of HRRP, but describes the statistical characteristics of each HRRP frame of a target and regards the frame model with the maximum class-conditional probability density as the target’s statistical model, which is referred to as frame model approach corresponding to mixture model approach in this paper.

B. Time-Shift Sensitivity

There are two kinds of time-shift sensitivity in HRRP-based statistical recognition.

- Time-shift sensitivity in training phase:* Each frame of training samples should be time-shift compensated,

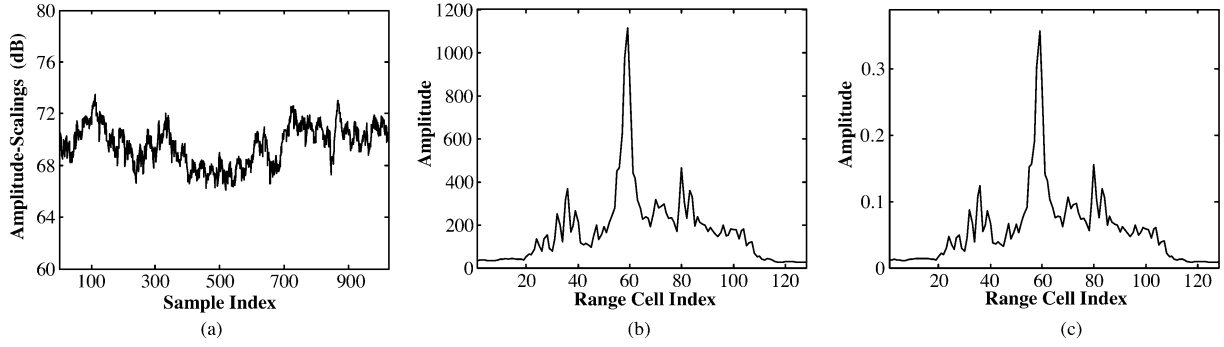


Fig. 1. An analysis of the amplitude-scale sensitivity of an HRRP frame in “Yark-42” measured data. (a) The amplitude-scales of an HRRP frame. (b) The average profile of an original HRRP frame. (c) The average profile of an amplitude normalized HRRP frame.

which can be achieved by using the time-shift compensation techniques in ISAR imaging, such as the slide correlation processing or its modifications [21].

- b) *Time-shift sensitivity in test phase:* In theory, when the posterior probability of a time-shift compensated test sample is the largest, the time-shift compensation in test phase is optimal. Since an HRRP includes many range cells, it requires large computation burden to search the optimal time-shift compensation by shifting the test sample between range cells one by one. Although the time-shift compensation obtained by the slide correlation processing is not optimal, its computation is small. The two time-shift compensation approaches will be compared based on measured data in Section VI

C. Amplitude-Scale Sensitivity

If training samples and test samples are measured under same conditions, the amplitude-scales of HRRPs, which contain the target size signatures, can be used in HRRP-based RATR. Actually, the measurement conditions are difficult to be kept exactly same in applications. Therefore, usually only the waveforms of HRRPs without amplitude-scale sensitivity can be used in HRRP target recognition. This is accomplished by amplitude normalization, e.g., L_2 normalization. In theory, the statistical characteristics of amplitude normalized HRRPs will be different from those of the original HRRPs. However, there is no variation in the distribution form of an HRRP frame but in the distribution parameters, because the amplitude-scales of an HRRP frame don't change much. This can be illustrated by Fig. 1 which comes from the measured data of “Yark-42” (detailed introduction shown in Section VI-A). Fig. 1(a) shows the amplitude-scales of an HRRP frame. Fig. 1(b) and (c) compares the two kinds of average profiles, i.e., the average profile of the original HRRP frame and that of the amplitude normalized HRRP frame, which can reflect the effect of amplitude normalization on the statistical characteristics of the HRRP frame. Obviously, the waveforms of the two average profiles are similar. By the amplitude normalization process, the amplitude-scale information is not available, but the test samples are comparable with the frame models, which is the basic prerequisite of HRRP-based statistical recognition. The remaining problem is how to accurately describe the statistical characteristics of each frame.

III. STATISTICAL MODELS FOR TARGET HRRP

A. Statistical Characteristics of Target HRRP

According to Section II-A, the immovable scatterers and their distribution in a range cell are irrelevant to those in another cell within the target-aspect sector corresponding to an HRRP frame, thus the range cells of an HRRP approximate to be statistically independent. Therefore, study on the statistical characteristics of target HRRP turns into study on those of the echoes per range cell within the target-aspect sector corresponding to an HRRP frame. According to the scattering center model, there are three kinds of scatterer distribution in a range cell.

- a) *First type of range cell:* There are a large number of small scatterers and no predominant scatterers in this type of range cell. Under the hypothesis that the intensities of the small scatterers are almost same and the number of the small scatterers is large enough, the central limit theorem holds. If the echoes' initial phases of the small scatterers are uniform-distributed, according to the central limit theorem, the inphase or quadrature component of the complex echo of this type of cell will follow Gaussian distribution with zero mean and its amplitude will follow Rayleigh distribution [23]. The probability density function (pdf) of Rayleigh distribution is

$$p_{\text{Rayleigh}}(x) = \begin{cases} \frac{x}{b^2} \exp\left(-\frac{x^2}{2b^2}\right) & x > 0 \\ 0 & x \leq 0 \end{cases} \quad (2)$$

where b is the scale parameter;

- b) *Second type of range cell:* This type of range cell consists of a predominant scatterer and a large number of small scatterers. Under the hypothesis similar to the first type, the inphase or quadrature component of the complex echo of this type of cell will follow Gaussian distribution with the mean determined by the echo's inphase or quadrature component of the predominant scatterer, in accordance with the central limit theorem. Thus, its amplitude follows a Ricean distribution amplitude [23]. The pdf of Ricean distribution is

$$p_{\text{Rice}}(x) = \begin{cases} \frac{x}{b^2} \exp\left(-\frac{x^2+|v|^2}{2b^2}\right) I_0\left(\frac{x|v|}{b^2}\right) & x > 0 \\ 0 & x \leq 0 \end{cases} \quad (3)$$

where b is the scale parameter, v is the noncentrality parameter, and $I_0(\cdot)$ is the modified Bessel function of zero order

$$I_0(c) = \sum_{k=0}^{\infty} \left(\left(\frac{c^2}{4} \right)^k / (k! \Gamma(k+1)) \right)$$

with $\Gamma(\cdot)$ denoting the Gamma function

$$\Gamma(c) = \int_0^{\infty} \exp(-u) u^{c-1} du.$$

- c) *Third type of range cell*: There are a large number of small scatterers and several predominant scatterers in this type of range cell. For the sake of simplicity, assume that there are only two predominant scatterers with their echoes' initial phases uniform-distributed in a range cell. If a target uniformly rotates within the target-aspect sector corresponding to an HRRP frame, the m th complex returned echo ($m = 0, 1, \dots, M-1$) in the n th range cell ($n = 1, 2, \dots, N$) is shown in (4) at the bottom of the page where $\Delta r_i(m) = y_{0i}(\cos(m \cdot \Delta\phi) - 1) + x_{0i} \sin(m \cdot \Delta\phi)$ represents the radial displacement of the i th scatterer in the m th returned echo (as shown in Fig. 2, $\Delta\phi$ denotes the target-aspect difference between two neighboring echoes, x_{0i} and y_{0i} represent the initial cross and radial coordinates of the i th scatterer), λ is the wavelength, σ_i and ψ_{0i} represent the echo's strength and initial phase of the i th scatterer, respectively. If $|x_{02} - x_{01}|$ is much larger than λ , namely, the cross range between the two scatterers is large enough compared with the radar signal wavelength (this condition can be met in many cases), the pdf of the echo amplitude will have two peaks at $\sigma_1 + \sigma_2$ and $|\sigma_1 - \sigma_2|$. If there are more than two predominant scatterers in one range cell, its statistical property can be approximately represented by a multimodal (mainly bimodal) distribution, which depends on the particular geometrical distribution of the predominant scatterers. Actually, there are many cells' echoes following multimodal distributions in our measured data (detailed analysis shown in Section VI-B). However, [5], [18], and [19] did not take such distribution into consideration. This paper will introduce the multimodal distribution into HRRP-based statistical recognition.

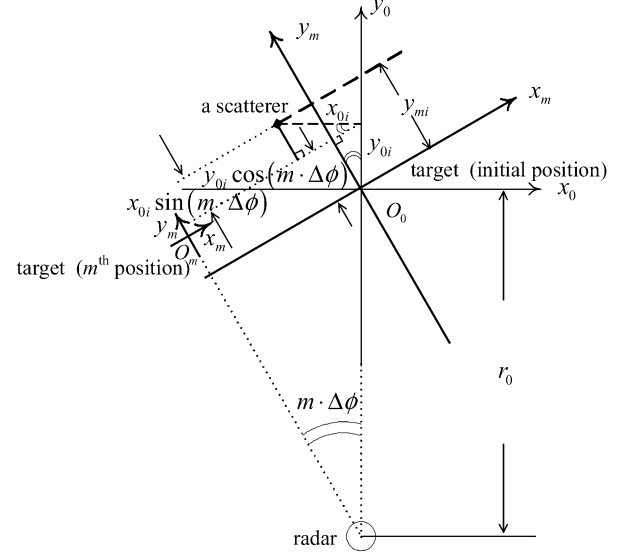


Fig. 2. Target rotation sketch map.

One point worth paying attention to is that the above analysis, i.e., analysis on the statistical characteristics of original HRRPs, is also suitable for amplitude normalized HRRPs, because amplitude normalization hasn't much effect on the distribution form of an HRRP frame.

B. Statistical Description of Target HRRP

As aforementioned in Section I, accurate description of HRRPs statistical characteristics is a great challenge in HRRP-based statistical recognition. Now there are two statistical models, i.e., Gaussian model [5] and Gamma model [18], [19], available for target HRRP. Gaussian model is based on that the power transformed echoes per range cell approximate to be Gaussian-distributed [5], [6], [24]. Two problems arise when Gaussian model is used. First, the power parameter used in power transformation needs to be carefully selected in the training phase, otherwise, the recognition performance will be degraded seriously. Second, power transformed HRRPs are more sensitive to the signal-to-noise-ratio (SNR) than original HRRPs [7]. Gamma model is motivated by the fact that Gamma distribution has been empirically observed to model target cross-section fluctuations under the low-resolution condition [25]. However, as aforementioned, the statistical characteristics of target HRRP are much more complex than those of

$$\begin{aligned} |x_n(m)| &= \left| \sum_{i=1}^2 \sigma_i e^{-j[\frac{4\pi}{\lambda} \Delta r_i(m) - \psi_{0i}]} \right| \\ &= \left| \sigma_2 e^{-j[\frac{4\pi}{\lambda} \Delta r_1(m) - \psi_{01}]} \right| \cdot \left| \frac{\sigma_1}{\sigma_2} + e^{-j\{\frac{4\pi}{\lambda} [\Delta r_2(m) - \Delta r_1(m)] - (\psi_{02} - \psi_{01})\}} \right| \\ &= \sigma_2 \cdot \left| \frac{\sigma_1}{\sigma_2} + e^{-j\{\frac{4\pi}{\lambda} [(y_{02} - y_{01}) \cdot (\cos(m\Delta\phi) - 1) + (x_{02} - x_{01}) \cdot \sin(m\Delta\phi)]\}} \cdot e^{j(\psi_{02} - \psi_{01})} \right| \\ &\stackrel{y_{01} \approx y_{02}}{\approx} \sigma_2 \cdot \left| \frac{\sigma_1}{\sigma_2} + e^{-j\{\frac{4\pi}{\lambda} [(x_{02} - x_{01}) \cdot \sin(m\Delta\phi)]\}} \cdot e^{j(\psi_{02} - \psi_{01})} \right| \end{aligned} \quad (4)$$

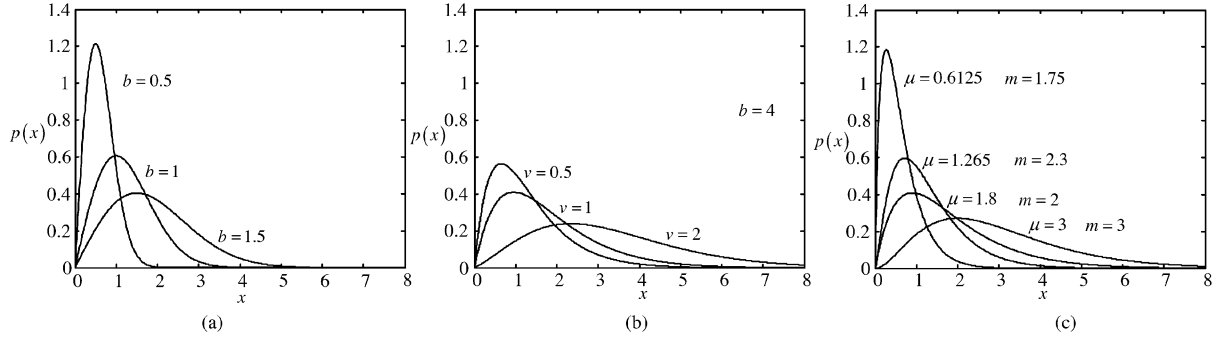


Fig. 3. The pdfs for some Rayleigh, Ricean, and Gamma variates. (a) Rayleigh distribution. (b) Ricean distribution. (c) Gamma distribution.

point-target's echoes with low resolution radar, e.g., the echoes of the third type of range cell following a multimodal distribution. Statistical description of the three types of echoes will be discussed in the following.

- a) Under the ideal condition, the echoes of the first and second types of range cells are Rayleigh-distributed and Ricean-distributed, respectively. What the ideal condition requires is a large number of the scatterers with almost same intensities per range cell, which ensures that the central limit theorem holds. However, increasing the resolution implies a reduction of the number of scatterers per range cell, thus the Rayleigh distribution and Ricean distribution tend to fail for the echoes of the first and second types of cells under the high-resolution condition. More flexible distribution forms should be selected. In this paper, the first and second types of echoes are modeled with Gamma distribution. The pdf of Gamma distribution is

$$p_r(x) = \begin{cases} \frac{m}{\mu \Gamma(m)} \left(\frac{mx}{\mu}\right)^{m-1} \exp\left(-\frac{mx}{\mu}\right) & x > 0 \\ 0, & x \leq 0 \end{cases} \quad (5)$$

where μ is the mean, m is the order parameter. The reasons why Gamma distribution is selected are as follows.

- i) The echo value per range cell is larger than zero, while a Gamma variant is also larger than zero.
 - ii) The pdfs of Gamma family are similar to those of Rayleigh distribution and Ricean distribution, all of which have “heavy tails.” The pdfs for some Rayleigh, Ricean, and Gamma variates are shown in Fig. 3.
 - iii) Gamma distribution is a flexible distribution with two parameters, i.e., mean parameter and order parameter. By changing the two parameters, Gamma family can approximate to Exponential distribution, χ^2 distribution, Gaussian distribution and so on [23].
 - iv) Compared to Ricean distribution and other complex distributions, Gamma distribution is simple, of which the two parameters can be estimated by the ML method.
- b) The echo of the third type of range cell follows a multimodal distribution. A mixture distribution can approximate such distribution. For the sake of simplicity, this paper selects Gaussian mixture distribution. Actually, the pdf of any small sample data set can be approximated by a Gaussian mixture distribution, so long as the number of

mixture components is large enough. The more mixture components the Gaussian mixture distribution has, the more precise the estimated pdf will be. However, too complex data representation model may result in overfitting and cannot guarantee good generalization performance, especially for small sample data set problem [26], [27]. Generally, for HRRP-based RATR, training data available is incomplete [8]–[10]. In this case, generalization performance is one of the most important factors worth paying attention to. To avoid overfitting, we limit the number of mixture components less than four in our experiments.

Actually it is difficult to model target HRRP as a simple statistical distribution. According to the above analysis, this paper proposed that echoes of different types of cells are modeled as the corresponding distribution forms, of which the first and second types of echoes can be modeled with Gamma distribution, while the third type with Gaussian mixture distribution. Such a two-distribution compounded statistical model can describe the statistical characteristics of HRRPs more accurately than Gaussian model and Gamma model.

IV. PARAMETER ESTIMATION

When a statistical model is available, the classifier training problem is turned into a parameter estimation problem, and we face the choice of an accurate estimation process.

A. Estimation for the Distribution Form

In order to model echoes of different types of range cells as the corresponding distribution forms, the type of each cell should be known at first. As aforementioned in Section III, as far as the composition is concerned, the numbers of predominant scatterers in the three types of cells are different; as far as the distribution form is concerned, the first and second types of echoes follow unimodal distributions, while the third follows a multimodal distribution. Therefore, the number of clusters of training data can reflect the type of a range cell, in other words, the first and second types of echoes should have one cluster, whereas the third should have several. Xu *et al.* proposed the RPCL algorithm that can perform appropriate clustering without knowing the number of clusters by automatically driving extra seed points far away from the input data set [22]. Given a set of data $\{\mathbf{y}_i\}_{i=1}^N$ ($\mathbf{y}_i$ is an M -dimensional vector) that forms J^* clusters (J^*

is unknown), the RPCL algorithm is to perform clustering by learning J seed points ($J > J^*$), denoted as

$$\{\omega_j^{(0)}\}_{j=1}^J$$

The RPCL algorithm mainly consists of the following two steps.

Assume that $t = 0$, $i = 1$, and the threshold Th is a small positive number.

Step 1) Let

$$I(j|\mathbf{y}_i) = \begin{cases} 1, & j = c \\ -1, & j = r \\ 0 & \text{otherwise} \end{cases} \quad 1 \leq j \leq J \quad (6)$$

with

$$\begin{aligned} c &= \arg \min_j \gamma_j \|\mathbf{y}_i - \omega_j\|^2 \\ r &= \arg \min_{j \neq c} \gamma_j \|\mathbf{y}_i - \omega_j\|^2 \end{aligned} \quad (7)$$

where $\gamma_j = (n_j / \sum_{r=1}^J n_r)$ is the relative winning frequency of the seed point ω_j in the past, and n_j is the cumulative number of occurrences of $I(j|\mathbf{y}_i) = 1$ in the past.

Step 2) Update the weight vector

$$\{\omega_j^{(t)}\}_{j=1}^J$$

by

$$\omega_j^{(t)}|_{\text{new}} = \omega_j^{(t)}|_{\text{old}} + \Delta\omega_j \quad (8)$$

with

$$\Delta\omega_j = \begin{cases} \alpha_c (\mathbf{y}_i - \omega_j^{(t)}) & I(j|\mathbf{y}_i) = 1 \\ -\alpha_r (\mathbf{y}_i - \omega_j^{(t)}) & I(j|\mathbf{y}_i) = -1 \\ 0 & \text{otherwise} \end{cases} \quad 1 \leq j \leq J \quad (9)$$

where $0 \leq \alpha_c, \alpha_r \leq 1$ are the learning rates for the winner and rival unit, and often set $\alpha_r \ll \alpha_c$ as shown in [22]. Let $i = i + 1$. If $i \leq N$, let the algorithm go back to Step 1).

If $i > N$ and $\|\Delta\omega_1^T, \Delta\omega_2^T, \dots, \Delta\omega_J^T\| > Th$, let $t = t + 1$, $i = 0$ and the algorithm go back to Step 1. If $i > N$ and $\|\Delta\omega_1^T, \Delta\omega_2^T, \dots, \Delta\omega_J^T\| \leq Th$, let the iteration end. According to the number of samples in each cluster, the cluster label is defined as

$$\begin{aligned} \text{Cluster_Index}(j) &= \begin{cases} 1 & \text{Num}\{I(j|\mathbf{y}_i) = 1 | i = 1, 2, \dots, N\} > 0 \\ 0 & \text{Num}\{I(j|\mathbf{y}_i) = 1 | i = 1, 2, \dots, N\} = 0 \end{cases} \\ &1 \leq j \leq J. \end{aligned} \quad (10)$$

Then the number of clusters after the learning is

$$N_C = \text{Num}\{\text{Cluster_Index}(j) = 1 | j = 1, 2, \dots, J\}. \quad (11)$$

Compared with conventional competitive learning (CL) algorithms, the basic idea of the RPCL algorithm is that for each input not only the weight vector ω_c of the unit which wins the competition is modified to adapt to the input, but also the weight vector ω_r of its rival, i.e., the second winner, is deleared by a learning rate smaller than that used by ω_c . Thus, the appropriate number of units will be selected automatically for representing an input data set by gradually driving extra units far away from the distribution of data set in the case that the number of units in a competitive learning net is larger than the number of clusters in the input data set. If echo data of a range cell have one cluster, we use Gamma distribution to approximate to the echoes' pdf; otherwise, we use Gaussian mixture distribution. As explained in Section III-B-b), to avoid overfitting, we assume that $J = 4$ in our experiments.

When the number of clusters per range cell's echo is estimated, the cell's type can be distinguished and its echo can be modeled as the corresponding distribution form.

B. Estimation for the Parameters of Gamma Distribution

Given a set of data $\{\mathbf{y}_i\}_{i=1}^N$ ($\mathbf{y}_i$ is an M -dimensional vector) of which each component $y_i(d)$ ($d = 1, 2, \dots, M$) is independent and follows Gamma distribution, the loglikelihood function is

$$\begin{aligned} L(\theta) &= \sum_{d=1}^M L(\theta(d)) = \sum_{d=1}^M \sum_{i=1}^N \ln p(y_i(d) | \theta(d)) \\ &= \sum_{d=1}^M \sum_{i=1}^N \ln \left[\frac{m(d)}{\mu(d) \cdot \Gamma(m(d))} \left(\frac{m(d) \cdot y_i(d)}{\mu(d)} \right)^{m(d)-1} \right. \\ &\quad \left. \times \exp \left(-\frac{m(d) \cdot y_i(d)}{\mu(d)} \right) \right] \end{aligned} \quad (12)$$

where $\theta = [\mu^T, m^T]^T$ denotes the unknown parameter set. The ML estimate of θ is given by

$$\theta_{ML} : \frac{\partial L(\theta)}{\partial \theta} = 0. \quad (13)$$

Substituting (12) into (13)

$$\mu_{ML}(d) = \frac{1}{N} \sum_{i=1}^N y_i(d) \quad d = 1, 2, \dots, M \quad (14)$$

and the m_{ML} satisfies

$$v(m_{ML}(d)) = -\frac{1}{N} \sum_{i=1}^N \ln \frac{y_i(d)}{\mu_{ML}(d)} \quad d = 1, 2, \dots, M \quad (15)$$

where $v(m_{ML}(d)) = \ln(m_{ML}(d)) - \psi(m_{ML}(d))$ and $\psi(\cdot)$ is the digamma function $\psi(c) = (d(\log(\Gamma(c)))/dc) = (d(\Gamma(c))/\Gamma(c))$. The function $\psi(c)$ satisfies the property that [28]

$$\psi^{(n)}(c) = (-1)^{n+1} n! \sum_{k=0}^{\infty} (c+k)^{-n-1} \quad c \neq 0, -1, -2, \dots \quad (16)$$

Thus

$$\begin{aligned} v'(m_{ML}(d)) &= \frac{1}{m_{ML}(d)} - \sum_{k=0}^{\infty} \frac{1}{(m_{ML}(d) + k)^2} \\ &= - \sum_{k=1}^{\infty} \frac{1}{(m_{ML}(d) + k)^2} < 0. \end{aligned} \quad (17)$$

Therefore, the function $v(m_{ML}(d))$ is monotonously decreasing for $m_{ML}(d) > 0$, and there must be a solution $m_{ML}(d) > 0$ for which $v(m_{ML}(d)) = \delta$ for $\delta > 0$. A simple gradient searching scheme (e.g., steepest descent method or Newton method) should be sufficient to obtain a solution.

C. Estimation for the Parameters of Gaussian Mixture Distribution

Given a set of data $\{\mathbf{y}_i\}_{i=1}^N$ ($\mathbf{y}_i$ is an M -dimensional vector) that follows Gaussian mixture distribution with J components, the pdf is

$$\begin{aligned} p(\mathbf{y}_i | \boldsymbol{\theta}) &= \sum_{j=1}^J P_j \cdot p(\mathbf{y}_i | \boldsymbol{\theta}_j) \\ &= \sum_{j=1}^J P_j \cdot \frac{1}{(2\pi)^{D/2} |\boldsymbol{\Sigma}_j|^{1/2}} \\ &\quad \times \exp \left\{ -\frac{1}{2} (\mathbf{y}_i - \boldsymbol{\mu}_j)^T \boldsymbol{\Sigma}_j^{-1} (\mathbf{y}_i - \boldsymbol{\mu}_j) \right\} \end{aligned} \quad (18)$$

with

$$P_j \geq 0 \text{ and } \sum_{j=1}^J P_j = 1 \quad (19)$$

where $\boldsymbol{\theta} = [\boldsymbol{\theta}_1^T, \boldsymbol{\theta}_2^T, \dots, \boldsymbol{\theta}_J^T]^T$ with $\boldsymbol{\theta}_j = [\boldsymbol{\mu}_j^T, \boldsymbol{\Lambda}_j^T]^T$ ($j = 1, 2, \dots, J$) denotes the Gausssian parameter set. For the case of diagonal covariance matrix, $\boldsymbol{\Lambda}_j = \text{diag}(\boldsymbol{\Sigma}_j) = [\sigma_j^2(1), \sigma_j^2(2), \dots, \sigma_j^2(M)]^T$ and P_j is the prior probability of the j th component. Then the loglikelihood function of the j th component is

$$\begin{aligned} L(\boldsymbol{\theta}_j) &= \sum_{i=1}^N \ln(P_j \cdot p(\mathbf{y}_i | \boldsymbol{\theta}_j)) \\ &= \sum_{i=1}^N \ln \left(P_j \cdot \frac{1}{(2\pi)^{M/2} |\boldsymbol{\Sigma}_j|^{1/2}} \right. \\ &\quad \left. \times \exp \left\{ -\frac{1}{2} (\mathbf{y}_i - \boldsymbol{\mu}_j)^T \boldsymbol{\Sigma}_j^{-1} (\mathbf{y}_i - \boldsymbol{\mu}_j) \right\} \right). \end{aligned} \quad (20)$$

Let $\mathbf{P} = [P_1, P_2, \dots, P_J]^T$. The unknown parameter set for Gaussian mixture distribution is $\boldsymbol{\Theta} = [\boldsymbol{\theta}^T, \mathbf{P}^T]^T$. Due to the lack of information concerning the labels of $\{\mathbf{y}_i\}_{i=1}^N$, the typical ML estimation similar to (13) encounters difficulty. The missing label information makes our current problem a typical task with an incomplete data set. Under the circumstances, this paper selects the so-called EM algorithm, which has attracted a great

deal of interest over the past few years in a wide range of application involving tasks with incomplete data sets [26]. The EM algorithm mainly consists of the following two steps.

Assume that $t = 0$, $\boldsymbol{\Theta}(0)$ denotes the initial values of the unknown parameters, and the threshold ε is a small positive number.

E-Step The expectation of (20) is

$$\begin{aligned} Q(\boldsymbol{\Theta}; \boldsymbol{\Theta}(t)) &= E \left[\sum_{i=1}^N \ln(p(\mathbf{y}_i | \boldsymbol{\theta}_j) P_j) \right] \\ &= \sum_{i=1}^N \sum_{j=1}^J P(j | \mathbf{y}_i; \boldsymbol{\Theta}(t)) \ln(p(\mathbf{y}_i | \boldsymbol{\theta}_j) P_j) \end{aligned} \quad (21)$$

with

$$\begin{aligned} P(j | \mathbf{y}_i; \boldsymbol{\Theta}(t)) &= \frac{p(\mathbf{y}_i | \boldsymbol{\theta}_j(t)) P_j(t)}{p(\mathbf{y}_i; \boldsymbol{\Theta}(t))} \\ &= \frac{p(\mathbf{y}_i | \boldsymbol{\theta}_j(t)) P_j(t)}{\sum_{j=1}^J p(\mathbf{y}_i | \boldsymbol{\theta}_j(t)) P_j(t)} \quad 1 \leq j \leq J \end{aligned} \quad (22)$$

M-Step Maximizing (21) with respect to $\boldsymbol{\theta}$ and $\mathbf{P}$. The ML estimate of $\boldsymbol{\Theta}(t+1)$ is given by

$$\begin{aligned} \boldsymbol{\theta}_{ML}(t+1) : \frac{\partial Q(\boldsymbol{\Theta}; \boldsymbol{\Theta}(t))}{\partial \boldsymbol{\theta}} &= \mathbf{0} \\ \mathbf{P}_{ML}(t+1) : \begin{cases} \frac{\partial R(\mathbf{P}; \mathbf{P}(t))}{\partial \mathbf{P}} = \frac{\partial \left[Q(\boldsymbol{\Theta}; \boldsymbol{\Theta}(t)) + \lambda \left(1 - \sum_{j=1}^J P_j \right) \right]}{\partial \mathbf{P}} = \mathbf{0} \\ \text{s.t. } \sum_{j=1}^J P_j = 1. \end{cases} \end{aligned} \quad (23)$$

Since $\sum_{j=1}^J P_j = 1$, the ML estimation of $\mathbf{P}$ is achieved by using a Langrangian multiplier. Substituting (21) into (23), we get (24) as shown at the bottom of the next page, where $\circ$ denotes dot product between vectors.

If $\|\boldsymbol{\Theta}(t+1) - \boldsymbol{\Theta}(t)\| > \varepsilon$, let $t = t+1$ and the algorithm go back to E-step. Otherwise, let $\boldsymbol{\Theta}_{EM} = \boldsymbol{\Theta}_{ML}(t+1)$ and the iteration end.

So far, we can build a two-distribution compounded statistical model to represent the statistical characteristics of target HRRP.

V. MAIN PROCEDURE OF RADAR HRRP TARGET RECOGNITION BASED ON THE PROPOSED TWO-DISTRIBUTION COMPOUNDED STATISTICAL MODEL

The main procedure of radar HRRP target recognition based on the proposed statistical model will be described in the following. Nevertheless, a fact worth pointing out is that the procedure does not include the time-shift compensation of test sample, since two time-shift compensation approaches will be applied to the recognition experiments in Section VI and they have different test processes, which will be discussed in Section VI-D in detail.

A. Training Phase

- Step 1) Divide the training samples of target T_k ($k = 1, 2, \dots, K$) into HRRP frames $\{\mathbf{X}_l^{(k)}\}_{l=1}^L$, where $\mathbf{X}_l^{(k)} = \{\mathbf{x}_{li}^{(k)} \mid i = 1, 2, \dots, N\}$ denotes the l th range aligned and amplitude normalized HRRP frame.
- Step 2) Estimate the number of clusters in each range cell of each HRRP frame, which is marked as $N_C|_{(K \times D \times L)}$ with D denoting the number of range cells, by using the RPCL algorithm introduced in Section IV-A, where $\{\mathbf{y}_i\}_{i=1}^N = \{\mathbf{x}_{li}^{(k)}(d)\}_{i=1}^N$ ($d = 1, 2, \dots, D; l = 1, 2, \dots, L; k = 1, 2, \dots, K$) and $M = 1$.
- Step 3) Distinguish the type of each range cell according to the number of clusters.

$$\text{Cell_Index}(k, d, l) = \begin{cases} 1, & N_C(k, d, l) = 1 \\ -1 & N_C(k, d, l) > 1 \end{cases} \quad \begin{matrix} 1 \leq k \leq K, \\ 1 \leq d \leq D \\ 1 \leq l \leq L. \end{matrix} \quad (25)$$

- Step 4) If $\text{Cell_Index}(k, d, l) = 1$, its echoes are modeled as Gamma distribution with the parameters $\{\mu, m \mid k, l, d\}$ estimated by (14) and (15), where $\{\mathbf{y}_i\}_{i=1}^N = \{\mathbf{x}_{li}^{(k)}(d)\}_{i=1}^N$ and $M = 1$; If $\text{Cell_Index}(k, d, l) = -1$, its echoes are modeled as Gaussian mixture distribution with $N_C(k, d, l)$ components and the parameters $\{\boldsymbol{\mu}, \mathbf{A}, \mathbf{P} \mid k, l, d\}$ estimated by the EM algorithm introduced in Section IV-C, where $\{\mathbf{y}_i\}_{i=1}^N = \{\mathbf{x}_{li}^{(k)}(d)\}_{i=1}^N$ and $M = 1$.
- Step 5) Store both the distribution form and the parameters of each range cell for each frame of each target.

B. Test Phase

- Step 1) Substitute the amplitude normalized test sample $\mathbf{x}$ into the l th ($l = 1, 2, \dots, L$) frame model of target T_k ($k = 1, 2, \dots, K$). See (26) at the bottom of the page. Under the hypothesis that the range cells are statistically independent, the joint distribution of the test sample can be written as

$$p(\mathbf{x} \mid l; k) = \prod_{d=1}^D p(x(d) \mid l; k) \quad 1 \leq k \leq K, \quad 1 \leq l \leq L \quad (27)$$

- Step 2) The class-conditional density of target T_k ($k = 1, 2, \dots, K$) is

$$p(\mathbf{x} \mid k) = \max_l p(\mathbf{x} \mid l; k). \quad (28)$$

According to Bayes theorem, the posterior probability of target T_k ($k = 1, 2, \dots, K$) is

$$p(k \mid \mathbf{x}) = \frac{p(\mathbf{x} \mid k) P(k)}{\sum_{k=1}^K p(\mathbf{x} \mid k) P(k)} \quad (29)$$

- Step 3) If

$$C = \arg \max_k p(k \mid \mathbf{x}) \quad (30)$$

$\mathbf{x}$ belongs to target T_C .

VI. EXPERIMENTAL RESULT

A. Measured Data

The results presented here are based on measured airplane data. The parameters of the targets and radar are shown in Table I, and the projections of target trajectories onto ground plane are shown in Fig. 4, from which the aspect angle of an

$$\left\{ \begin{array}{l} \mu_j(t+1) = \frac{\sum_{i=1}^N P(j|\mathbf{y}_i; \boldsymbol{\Theta}(t)) \mathbf{y}_i}{\sum_{i=1}^N P(j|\mathbf{y}_i; \boldsymbol{\Theta}(t))} \\ A_j(t+1) = \frac{\sum_{i=1}^N P(j|\mathbf{y}_i; \boldsymbol{\Theta}(t)) [\mathbf{y}_i - \mu_j(t+1)] [\mathbf{y}_i - \mu_j(t+1)]^T}{\sum_{i=1}^N P(j|\mathbf{y}_i; \boldsymbol{\Theta}(t))} \quad 1 \leq j \leq J \\ P_j(t+1) = \frac{1}{N} \sum_{i=1}^N P(j \mid \mathbf{y}_i; \boldsymbol{\Theta}(t)) \end{array} \right. \quad (24)$$

$$p(x(d) \mid l; k) = \begin{cases} \frac{m}{\mu \Gamma(m)} \left(\frac{m \cdot x(d)}{\mu} \right)^{m-1} \exp \left(-\frac{m \cdot x(d)}{\mu} \right) & \text{Cell_Index}(k, d, l) = 1 \\ \sum_{j=1}^{N_C(k, d, l)} P_j \cdot \frac{1}{\sqrt{2\pi}\sigma_j} \exp \left\{ -\frac{1}{2} \left(\frac{x(d) - \mu_j}{\sigma_j} \right)^2 \right\} & \text{Cell_Index}(k, d, l) = -1 \end{cases} \quad 1 \leq d \leq D. \quad (26)$$

TABLE I
PARAMETERS OF PLANES AND RADAR IN THE ISAR EXPERIMENT

radar parameters	center frequency		5520 MHz
	bandwidth		400 MHz
planes	length (m)	width (m)	height (m)
Yark-42	36.38	34.88	9.83
An-26	23.80	29.20	9.83
Cessna Citation S/II	14.40	15.90	

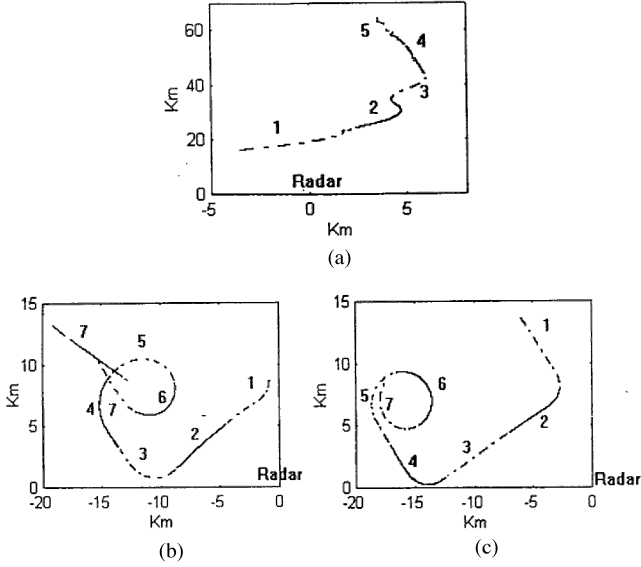


Fig. 4. The projections of target trajectories onto ground plane. (a) Yark-42. (b) An-26. (c) Cessna Citation S/II.

airplane can be estimated according to its relative position to radar. As shown in Fig. 4, the measured data are segmented. Training data and test data are from different data segments, among which the second and the fifth segments of Yark-42, the fifth and the sixth segments of An-26, the sixth and the seventh segments of Cessna Citation S/II are taken as the training data, other data segments are taken as the test data. As aforementioned, we need training data from different target-aspect angles to establish different frame models. This can be realized by using the simulated data generated by some electromagnetic simulation software, such as XPATCH, as the training data. However, for our measured data, we only require that the training data cover almost all of the target-aspect angles of the test data. According to Fig. 4, this can be satisfied. Moreover, the elevation angles of the test data are different from those of the training data, thus the generalization performance of the recognition methods can be tested. According to (1) and the radar parameters shown in Table I, and in view of few scatterers distributed at the edges of an aircraft-like target and with weak energy, HRRPs in about 3° target-aspect sector are taken as an HRRP frame in our experiments. With large SNR, the effect of noise in the measured data on the recognition can be ignored. In addition, the measured HRRP is a 128-dimensional vector.

B. Statistical Description of the Echoes of Some Typical Range Cells

Fig. 5 shows the approximated pdfs of the echoes of some typical range cells, where the bar represents that approximated by histogram method, the dotted line represents that by Parzen windows using Gaussian kernels and the solid line represents that by the statistical model proposed in this paper. Fig. 5(a)–(c) comes from the first and second types of cells whose echoes are modeled as Gamma distribution; Fig. 5(d)–(f) comes from the third type of cells whose echoes are modeled as Gaussian mixture distribution. Under the constraint $J = 4$, there are only two mixture components in the third type of cells in our experiment, and the PRCL only put the range cells with distinct bimodal distribution into the third type of range cell. The proportions of the third type of cells are 28.00%, 30.50%, and 47.00% for “Yark-42,” “Cessna Citation S/II,” and “An-26,” respectively.

C. Recognition Performances of the Three Statistical Models and Some Nonstatistical Methods

The average recognition rate of Gaussian model versus the power parameter is shown in Fig. 6, of which the optimal power parameter $\alpha^* = 0.5$. Tables II and III show the confusion matrices and average recognition rates of Gaussian model with the optimal power transformation ($\alpha^* = 0.5$), Gamma model, the two-distribution compounded statistical model, template matching method under the maximum correlation coefficient criterion (MCC-TMM) [10], MCC-TMM with the optimal power transformation ($\alpha^* = 0.1$) [6] and radial basis function network (RBFN) [10]. Obviously, statistical models have better recognition performances than the nonstatistical methods, and the average recognition rate of the two-distribution compounded statistical model is approximately 5 percentage points larger than those of Gaussian model and Gamma model, which are nearly equivalent.

According to the introduction in Section VI-A, the SNR in the measured data is large. For aircraft-like targets, the noises in the inphase and quadrature echoes of targets can be assumed to be Gaussian white noises; while for ship-like targets, those to be heavy-tailed and colored noises [29], i.e., sea clutter. In order to compare the effect of noises on the three statistical models, we simulate the two kinds of noises¹. The SNR is defined as

$$\text{SNR} = 10 \times \log_{10} \left(\frac{\bar{P}_x}{P_N} \right) = 10 \times \log_{10} \left(\frac{\sum_{d=1}^D P_{x(d)}}{D \cdot P_N} \right) \quad (31)$$

where $\bar{P}_x$ denotes the average power of original HRRP, $\{P_{x(d)}\}_{d=1}^D$ denotes the power of the original echo per range cell, D denotes the number of range cells, and P_N denotes the power of noise. Thus, $10^{\text{SNR}/10}$ equals to the rate of the average power of original HRRP to the power of the noise. The average recognition rates of the three models versus the SNR are shown in Fig. 7. Obviously, Gaussian model with power

¹Because we do not have measured ship data, we can only add some simulated heavy-tailed and colored noises data into our measured airplane data. Although it is not representative of a realistic case, it can show the behavior of the three statistical models when heavy-tailed and colored noises occur.

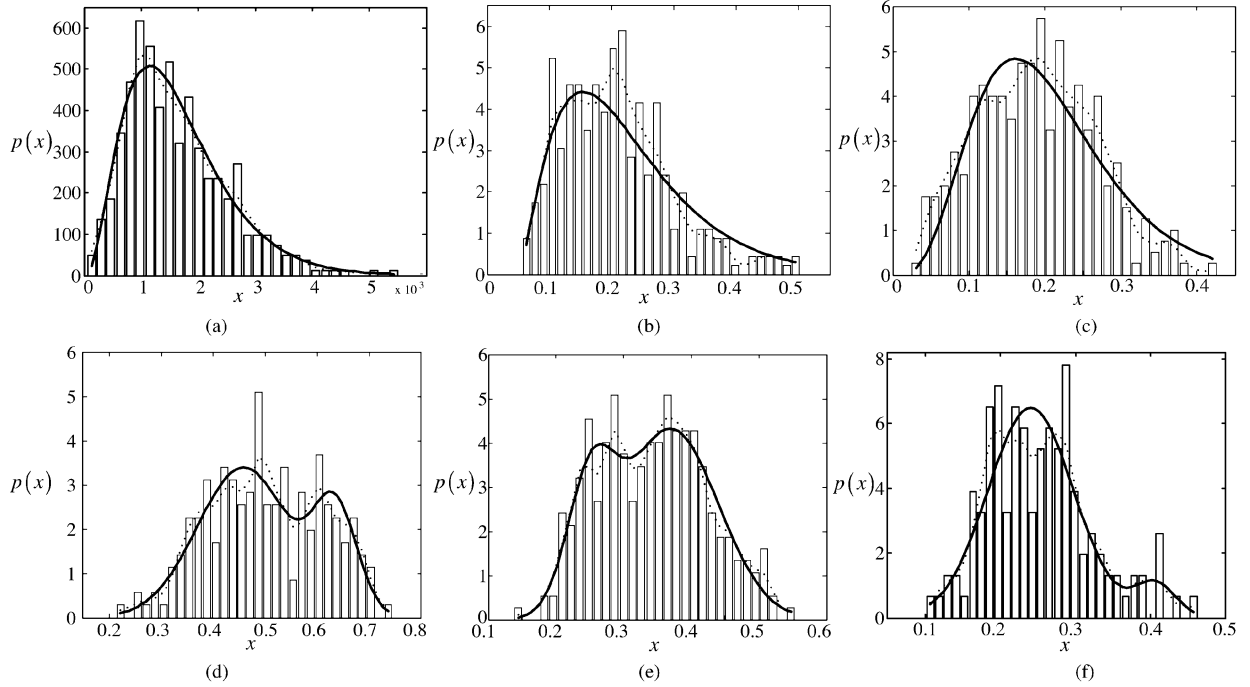


Fig. 5. The approximated pdfs of the echoes of some typical range cells: the bar represents the approximation of a pdf via histogram method; the dotted line represents the approximation of a pdf via Parzen windows using Gaussian kernels; the solid line represents the approximation of a pdf via our two-distribution compounded statistical model. (a) First type of resolution cell. (b) Second type of resolution cell. (c) Second type of resolution cell. (d) Third type of resolution cell. (e) Third type of resolution cell. (f) Third type of resolution cell.

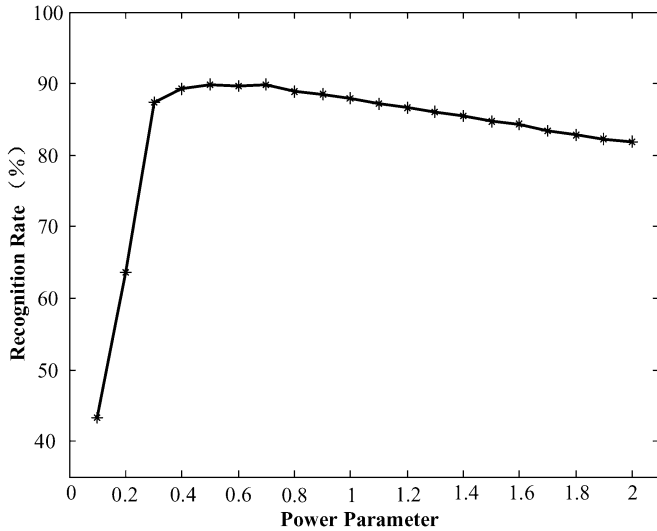


Fig. 6. The average recognition rate of Gaussian model versus the power parameter.

transformation is most sensitive to noises. With $\text{SNR} = 20$ dB, in Fig. 7(a), the average recognition rates of Gaussian model, Gamma model, and our two-distribution compounded statistical model are 60.67%, 70.67%, and 80.13%, respectively; while in Fig. 7(b), those are 62.37%, 70.74%, and 79.35%, respectively. Compared between Fig. 7(a) and (b), the recognition rate curves against heavy-tailed and colored noises are a little gentler. The reason comes from that heavy-tailed and colored noises follow the distribution with “heavy tail.” Given the same SNR, the Gaussian white noises distribute more “uniformly” in the range cells, while the “heavy-tailed” noises generally behaves as

several strong peaks and many weaker values in the range cells, thus the influence of the sea clutter on the posterior probability of the range cells mainly exhibits in the range cells with strong clutter. Considering that the recognition algorithm used in our manuscript is actually based on range cells fusion, therefore, the variation of recognition performance with SNR is a little gentler for heavy-tailed and colored noises than that for Gaussian white noises.

D. A Comparison Between the Two Time-Shift Compensation Approaches

In the above recognition experiments, for Gamma model and the proposed statistical model the test samples are time-shift compensated with each frame model by the slide correlation processing [21], whereas for Gaussian model by the maximum posterior probability approach. As mentioned in Section II-B, in theory, we should use the maximum posterior probability approach to time-shift compensation in test phase for the recognition based on the maximum posterior probability. For Gaussian model, we can obtain the fast algorithm of the maximum posterior probability approach to time-shift compensation, which is similar to the slide correlation algorithm, by simple formula derivation (due to the limitation on the length of this paper, we do not derive the fast algorithm in detail). However, there is not a similar fast time-shift compensation algorithm for Gamma model or the two-distribution compounded statistical model. In order to search the optimal time-shift compensation, we should shift the test sample between range cells one by one and select the time-shift compensation with the maximum posterior probability. The recognition performances of Gamma model and our

TABLE II
THE CONFUSION MATRICES AND AVERAGE RECOGNITION RATES OF GAUSSIAN MODEL WITH THE OPTIMAL POWER TRANSFORMATION, GAMMA MODEL AND THE TWO-DISTRIBUTION COMPOUNDED STATISTICAL MODEL

	Gaussian model ($\alpha^* = 0.5$)			Gamma model			the two-distribution compounded statistical model		
	Yark-42	Cessna Citation S/II	An-26	Yark-42	Cessna Citation S/II	An-26	Yark-42	Cessna Citation S/II	An-26
Yark-42	99.50	1.05	16.05	99.25	0.70	17.90	99.75	0	8.00
Cessna Citation S/II	0	96.00	9.95	0	89.50	1.40	0	93.60	1.60
An-26	0.50	2.95	74.00	0.75	9.80	80.70	0.25	6.40	90.40
average recognition rates P_{cc} (%)	89.83			89.82			94.58		

TABLE III
THE CONFUSION MATRICES AND AVERAGE RECOGNITION RATES OF MCC-TMM, MCC-TMM WITH THE OPTIMAL POWER TRANSFORMATION AND RBFN

	MCC-TMM			MCC-TMM ($\alpha^* = 0.1$)			RBFN		
	Yark-42	Cessna Citation S/II	An-26	Yark-42	Cessna Citation S/II	An-26	Yark-42	Cessna Citation S/II	An-26
Yark-42	81.25	1.00	14.75	98.42	0	12.45	76.25	0.50	10.25
Cessna Citation S/II	2.75	77.75	43.00	0.08	98.20	13.70	0	73.75	21.75
An-26	16.00	21.25	42.25	1.50	1.80	73.85	23.75	25.75	68.00
average recognition rates P_{cc} (%)	67.08			88.88			72.67		

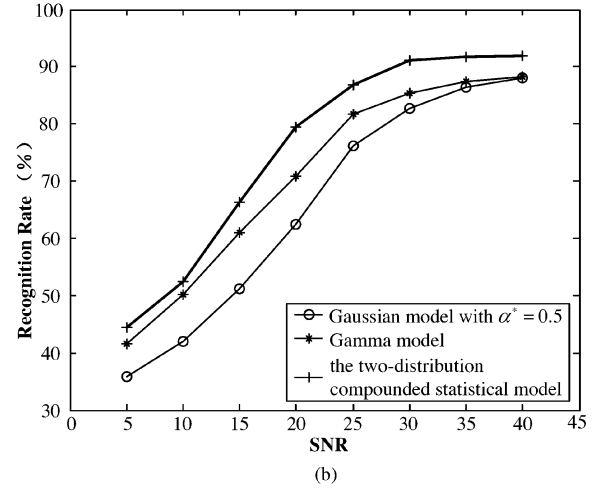
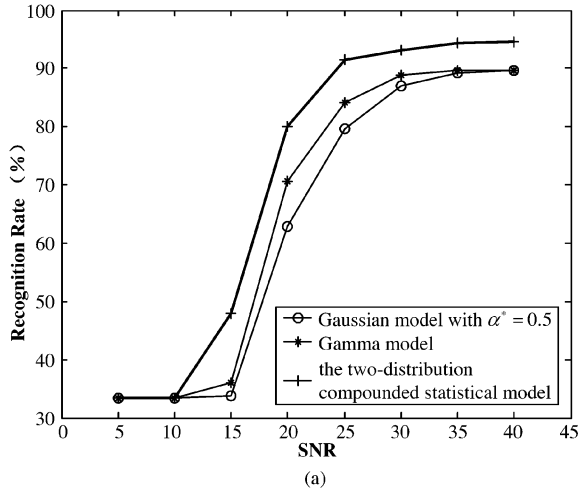


Fig. 7. The average recognition rates of Gaussian model, Gamma model and the two-distribution compounded statistical model versus the SNR. (a) Gaussian white noises. (b) Heavy-tailed and colored noises.

statistical model with the time-shift compensation by the maximum posterior probability approach are shown in Table IV. Compared with Table II, the average recognition rates obtained by the maximum posterior probability approach to time-shift compensation are larger than those by the slide correlation processing by approximately 2 percentage points.

However, since an HRRP includes many range cells, it requires large computation burden to search the optimal time-shift compensation by the maximum posterior probability approach for Gamma model or our statistical model. If D denotes the number of range cells (as introduced in Section VI-A, for our measured HRRP $D = 128$), the computation of the maximum posterior probability approach to time-shift compensation approximates to be D times larger than that of the slide correlation processing.

E. An Evaluation of the Computation Costs of the Three Statistical Models

For an off-line recognition system, the computation cost in the training phase can be ignored. Accordingly, we should only evaluate a recognition method's computation cost in the test phase. Compared with the Gaussian model, it is much more complex to calculate the pdf of Gamma distribution or Gaussian mixture distribution than that of Gaussian distribution. However, it is also difficult to evaluate the difference between these computation costs in quantity. Therefore, we give the time costs to match a test sample with all frame models in the three statistical models by using MATLAB and under completely same condition. The results are shown in Table V. Obviously, the computation cost of the proposed two-distribution compounded

TABLE IV
THE CONFUSION MATRICES AND AVERAGE RECOGNITION RATES OF GAMMA MODEL AND THE TWO-DISTRIBUTION COMPOUNDED STATISTICAL MODEL WITH TIME-SHIFT COMPENSATION BY THE MAXIMUM POSTERIOR PROBABILITY APPROACH

	Gamma model			the two-distribution compounded statistical model		
	Yark-42	Cessna Citation S/II	An-26	Yark-42	Cessna Citation S/II	An-26
Yark-42	99.25	0.30	12.25	100.00	0.80	6.40
Cessna Citation S/II	0.25	91.65	4.40	0	94.40	0.80
An-26	0.50	8.05	83.35	0	4.80	92.80
average recognition rates P_{cc} (%)	91.42			95.73		

TABLE V
THE TIME COSTS TO MATCH A TEST SAMPLE WITH ALL FRAME MODELS IN THE THREE STATISTICAL MODELS BY USING MATLAB AND UNDER COMPLETELY SAME CONDITION

statistical model	elapsed time (s)
the two-distribution compounded model	0.3120
Gamma model	0.1880
Gaussian model	0.1410

model is more than 2 times larger than that of Gaussian model. However, some techniques, such as chart checking for calculating Gamma function values, can be used to further decrease the computation burden in applications.

VII. SUMMARY AND DISCUSSION

There are two statistical models available for target HRRP, i.e., Gaussian model and Gamma model. After resolving the three sensitivity problems, i.e., the target-aspect, time-shift, and amplitude-scale sensitivity of HRRP, respectively, this paper develops a new statistical model comprising two distribution forms, i.e., Gamma distribution and Gaussian mixture distribution, by which echoes of different types of range cells are modeled as the corresponding distribution forms. Determination of the type of a range cell is achieved by using the RPCL algorithm, while estimation for the parameters of Gamma distribution and Gaussian mixture distribution by the ML method and the EM algorithm, respectively. Experimental results for measured data show that the proposed statistical model not only has better recognition performance but also is more robust to noises than Gaussian model and Gamma model. Furthermore, since this paper adopts frame model approach, the recognition performance can be further improved by using target-aspect information of test samples, as shown in [8].

It is worth pointing out that for radar HRRP-based statistical recognition, a large amount of training data are required to learn the statistical characteristics of target HRRP, therefore, the performance evaluation results will be sounder if more targets and more training data are concerned.

ACKNOWLEDGMENT

The authors wish to thank Associate Editor Prof. Y. Abramovich and the three anonymous reviewers for their valuable comments and suggestions, which have helped to significantly improve this manuscript.

REFERENCES

- [1] R. A. Mitchell and R. Dewart, "Overview of high range resolution radar target identification," presented at the Automatic Target Recognition Working Group, Monterey, CA, 1994.
- [2] H.-J. Li and S.-H. Yang, "Using range profiles as feature vectors to identify aerospace objects," *IEEE Trans. Antennas Propag.*, vol. 41, no. 3, pp. 261–268, 1993.
- [3] A. Zyweck and R. E. Bogner, "Radar target classification of commercial aircraft," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 32, no. 2, pp. 598–606, 1996.
- [4] R. A. Mitchell, "Robust high range resolution radar target identification using a statistic feature based classifier with feature level fusion," Ph.D., Univ. Dayton, Dayton, OH, 1997.
- [5] S. P. Jacobs, "Automatic target recognition using high-resolution radar range profiles," Ph.D., Washington Univ., St. Louis, MO, 1999.
- [6] R. Van der Heiden and F. C. A. Groen, "The Box-Cox metric for nearest neighbor classification improvement," *Pattern Recog.*, vol. 30, no. 2, pp. 273–279, 1997.
- [7] A. K. Shaw, R. Vagist, and R. Williams, "HRR-ATR using eigen-templates with observation in unknown target scenario," *Proc. SPIE*, vol. 4053, pp. 467–478, 2000.
- [8] R. Williams, J. Westerkamp, D. Gross, and A. Palomino, "Automatic target recognition of time critical moving targets 1D high range resolution (HRR) radar," *IEEE Aerosp. Electron. Syst. Mag.*, vol. 15, no. 4, pp. 37–43, 2000.
- [9] X.-J. Liao, Z. Bao, and M.-D. Xing, "On the aspect sensitivity of high resolution range profiles and its reduction methods," in *Rec. IEEE 2000 Int. Radar Conf.*, Alexandria, VA, May 2000, pp. 310–315.
- [10] L. Du, H.-W. Liu, and Z. Bao, "Radar HRRP target recognition by the higher-order spectra features," in *Proc. IASTED Int. Conf. Appl. Informatics*, Innsbruck, Austria, 2004, pp. 627–632.
- [11] K. T. Kim, D. K. Seo, and H. T. Kim, "Efficient radar target recognition using the MUSIC algorithm and invariant feature," *IEEE Trans. Antennas Propag.*, vol. 50, no. 3, pp. 325–337, 2002.
- [12] J. Zwart, R. Heiden, S. Gelsema, and F. Groen, "Fast translation invariant classification of HRR range profiles in a zero phase representation," *Inst. Elect. Eng. Proc.—Radar Sonar Navig.*, vol. 150, no. 6, pp. 411–418, 2003.
- [13] K. Rosenbach and J. Schiller, "Non co-operative air target identification using radar imagery: Identification rate as a function of signal bandwidth," in *Rec. IEEE 2000 Int. Radar Conf.*, Alexandria, VA, May 2000, pp. 305–309.
- [14] D. E. Nelson, J. A. Starzyk, and D. D. Ensley, "Iterated wavelet transformation and signal discrimination for HRR radar target recognition," *IEEE Trans. Syst., Man, Cybern. A, Syst., Humans*, vol. 33, no. 1, pp. 52–57, 2002.
- [15] X.-J. Liao, P. Runkle, and L. Carin, "Identification of ground targets from sequential high-range-resolution radar signatures," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 38, no. 4, pp. 1230–1242, 2002.
- [16] H.-W. Liu and Z. Bao, *Radar HRR Profiles Recognition Based on SVM With Power-Transformed-Correlation Kernel*. Berlin, Germany: Springer-Verlag, 2004, Springer Lecture Notes in Computer Science, pp. 531–536.
- [17] Y. Shi and X.-D. Zhang, "A Gabor atom network for signal classification with application in radar target recognition," *IEEE Trans. Signal Process.*, vol. 49, no. 12, pp. 2994–3004, Dec. 2001.
- [18] A. R. Webb, "Gamma mixture models for target recognition," *Pattern Recog.*, vol. 33, pp. 2045–2054, 2000.

- [19] K. Copsey and A. R. Webb, "Bayesian gamma mixture model approach to radar target recognition," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 39, no. 4, pp. 1201–1217, Oct. 2003.
- [20] W. G. Carrara, R. S. Goodman, and R. M. Majewski, *Spotlight Synthetic Aperture Radar—Signal Processing Algorithms*. Norwood, MA: Artech House, 1995.
- [21] W. Ye, "Study of the inverse synthetic aperture radar imaging and motion compensation," Ph.D., Xidian Univ., Xi'an, China, 1996.
- [22] L. Xu, A. Krzyzak, and E. Oja, "Rival penalized competitive learning for clustering analysis, RBF net, and curve detection," *IEEE Trans. Neural Netw.*, vol. 4, no. 4, pp. 636–648, Jul. 1993.
- [23] M. Evans, N. Hastings, and B. Peacock, *Statistical Distributions*, Second ed. New York: Wiley, 1993.
- [24] K. Fukunaga, *Introduction to Statistical Pattern Recognition*, (2nd ed. Boston, MA: Academic, 1990.
- [25] M. Skolnik, *Introduction to Radar Systems*, 2nd ed. New York: McGraw-Hill, 1980.
- [26] R. O. Duda, P. E. Hart, and D. G. Stork, *Pattern Classification*, 2nd ed. New York: Wiley, 2001.
- [27] V. N. Vapnik, *The Nature of Statistical Learning Theory*. Berlin, Germany: Springer-Verlag, 1995.
- [28] M. Abramowitz and I. A. Stegun, *Handbook of Mathematical Functions with Formulas, Graphs, and Mathematical Tables*. New York: Dover, 1972.
- [29] W. Stehwien, "Statistics and correlation properties of high resolution X-band sea clutter," in *Rec. 1994 IEEE National Radar Conf.*, Atlanta, GA, Mar. 1994, pp. 46–51.



Lan Du received the B.Eng. and M.Eng. degrees in electronic engineering from Xidian University, Xi'an, China, in 2001 and 2004, respectively.

She is currently working toward the Ph.D. degree with the National Laboratory of Radar Signal Processing, Xidian University. Her research interests are in the fields of radar automatic target recognition (RATR), radar imaging (SAR/ISAR), and radar signal processing.



Hongwei Liu (M'00) received the M.Eng. and Ph.D. degrees in electronic engineering from Xidian University, Xi'an, China, in 1995 and 1999, respectively.

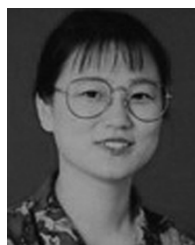
He is currently a Professor with the National Laboratory of Radar Signal Processing, Xidian University. From 2001 to 2002, he was a visiting scholar with the Department of Electrical and Computer Engineering, Duke University, Durham, NC. His research interests are radar automatic target recognition, radar signal processing, and adaptive signal processing.



Zheng Bao (M'80–SM'90) graduated from the Communication Engineering Institution of China, Xi'an, in 1953.

Currently, he is a Professor with Xidian University, Xi'an. He is the author or coauthor of six books and has published more than 300 papers. His research work focuses on the areas of space-time adaptive processing (STAP), radar imaging (SAR/ISAR), radar automatic target recognition (RATR), over-the-horizon radar (OTHR) signal processing, and passive coherent location (PCL).

Prof. Bao is a member of the Chinese Academy of Science.



Junying Zhang (M'02) received the Ph.D. degree in signal and information processing from Xidian University, Xi'an, China, in 1998.

From 2001 to 2002, she was a visiting scholar with the Department of Electrical Engineering and Computer Science, the Catholic University, Washington, DC. She is currently a Professor with the School of Computer Science and Engineering, and a part-time professor with the National Laboratory of Radar Signal Processing, Xidian University. Her research interests focus on intelligent information processing,

including automatic target recognition, cancer-related bioinformatics, medical and molecular imaging, machine learning and its application to cancer research, image processing, and pattern recognition.